How Far Does Translated-Query Fine-tuning Close the Arabic→English Retrieval Gap? An Empirical Study with an MTEB Arabic Benchmark and a Contamination Audit

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Abstract

Multilingual embedding models are clearly weaker when an Arabic query must find an English passage than in either monolingual direction. We measure how far fine-tuning on LLM-translated queries closes this gap and what it costs; the contributions are measurements, not a new method. (i) We assemble MTEB(ara, v1), a 19-task Arabic slice of MTEB, and audit its overlap with training data: 408 of the 1,000 MIRACL Arabic dev queries are near-verbatim TyDi QA training questions. (ii) On a held-out benchmark the ar→en MRR@10 of bge-m3 rises from 63.3 to 73.5 after fine-tuning and weight interpolation; we give the data-scaling curve and the human-versus-translated query offset. (iii) A table of negative results lists what did not help, with its dev effect. (iv) A WiSE trade-off curve shows the price paid on MIRACL and Mr.TyDi re-ranking. On MTEB(ara) the fine-tuned model is statistically indistinguishable from bge-m3 (mean difference +0.06, 95% task-bootstrap interval −1.1 to +1.1); seeds trained after the recipe was frozen give ar→en [TODO: pending]. (v) We also report CPU deployment costs of ONNX, int8 and weight-only 8-bit variants.

1 Introduction

Retrieval-augmented systems for Arabic users routinely search documents written in both Arabic and English. A multilingual bi-encoder should make this a non-issue, yet published multilingual models are markedly asymmetric: in our benchmark bge-m3 scores 87.8 for English-to-English retrieval but 63.3 when the query is Arabic and the passage English, and 67.8 in the opposite direction. A first version of this work (v1) fine-tuned a 118M multilingual-E5-small and lifted en→ar from 61.7 to 80.4 (above bge-m3's 67.8), but left ar→en at 55.1.

This is an empirical study. The recipe it evaluates (translated queries, mined hard negatives, a cached contrastive loss, distillation, weight interpolation) is assembled from published components, and we claim no method novelty. Our contributions are: (i) **MTEB(ara, v1) and a contamination audit**: a 19-task benchmark scored for 10 models, and a finding that one of its retrieval tasks overlaps heavily with a common training set; (ii) **the asymmetry and how far translated-query fine-tuning closes it**, including a data-scaling curve and the offset between machine-translated and human queries; (iii) a **negative-results table** of changes that did not help, each with its dev effect; (iv) the **WiSE trade-off curve** between cross-lingual gain and forgetting on guard tasks; (v) **deployment costs** of ONNX, int8 and weight-only 8-bit variants on CPU. We also report seeds trained after the recipe was frozen as the confirmatory result, and give the pre-registration trail in an appendix.

2 Related Work

Benchmarks. MTEB (Muennighoff et al., 2023) and its massively multilingual successor MMTEB (Enevoldsen et al., 2025) define the task suites in which our Arabic benchmark is expressed. Arabic retrieval is covered by MIRACL (Zhang et al., 2023), Mr.TyDi (Zhang et al., 2021), built on TyDi QA (Clark et al., 2020), and by the parallel QA sets XQuAD (Artetxe et al., 2020) and MLQA (Lewis et al., 2020). **Models.** Multilingual E5 (Wang et al., 2024, 2022) and M3-Embedding (bge-m3) (Chen et al., 2024) are the strong open multilingual encoders we start from. **Training components we reuse.** In-batch contrastive training (Henderson et al., 2017; Reimers and Gurevych, 2019) with gradient caching (Gao et al., 2021), mined hard negatives and their false-negative problem (Xiong et al., 2020; Qu et al., 2021), embedding-level distillation (Hinton et al., 2015; Hofstätter et al., 2020), and interpolation of fine-tuned and original weights, WiSE-FT (Wortsman et al., 2021), which we use

as a probe of the trade-off between gain and forgetting.

3 Benchmarks, Data and the Contamination Audit

Retrieval benchmark. The v1 benchmark uses only public data: TyDi QA GoldP, ARCD, SQuAD v1.1 (Rajpurkar et al., 2016; Mozannar et al., 2019) and XQuAD, split into article-disjoint train, dev and test pools. The test split has 6,823 queries over 9,817 passages; XQuAD's parallel questions give four directions (ar→ar, en→en, ar→en, en→ar) with identical information needs. Each slice is searched in the passages of its own pool in the passage language; we report MRR@10 per direction and the macro average of the four directions.

Translated queries. For cross-lingual training pairs, *questions only* are translated by a locally served LLM (Gemma-4-26B-A4B) into the other language; the passage keeps its language and stays the positive. v1 used 4,499 such pairs; v2 adds 20,999 from questions whose passages v1 never paired across languages. Test queries are never machine-translated, but the dev cross-lingual slices are, so absolute dev numbers are not comparable with test numbers. Translated XQuAD ar→en queries score 5.1–6.2 points higher than the human questions on the same 1,190 items for every model we tried: a near-constant offset, not evidence of an over-fit translation style, but a reason to read dev numbers only relatively.

MTEB(ara, v1). The benchmark has 19 tasks (retrieval, re-ranking, STS, pair classification, classification, clustering, bitext mining) with Arabic data, scored for 10 models (Table 8); the task selection and per-task licences are documented with the benchmark.

Contamination audit. While auditing the evaluation sets we found that 408 of the 1,000 Arabic dev queries of MIRACL-HardNegatives (41%) are near-verbatim questions (token Jaccard ≥ 0.8) of the TyDi QA Arabic *training* set, which v1 trained on. Against the full TyDi-ar and MIRACL-ar training questions the number is 816, leaving 184 clean queries. v2 therefore drops every training question that is a near-duplicate of a query of any audited task (7 MTEB tasks) and every one whose positive passage overlaps an evaluation positive; afterwards 0 near-duplicate MIRACL queries remain (Table 1).

Training queries	built	dropped	kept
Human questions (TyDi, ARCD, SQuAD)	16,419	1,173	15,246
Translated, v1	4,499	254	4,245
Translated, new in v2	20,999	1,113	19,886
Total (v2 recipe)	41,917	2,540	39,377

Table 1: Training queries of the v2 recipe before and after the contamination filter.

4 Models and Selection Protocol

We evaluate one recipe at three sizes; it is described for completeness, not as a contribution. All models minimise an InfoNCE loss over in-batch passages plus one mined hard negative per query, taken from ranks 1–30 of a baseline retriever in the same language pool, keeping only candidates at least 0.02 cosine below the positive. **v2-m3** fine-tunes bge-m3 with a cached multiple-negatives loss (Gao et al., 2021) (batch 64, mini-batch 16, scale 50, learning rate 1e-5, one epoch, maximum length 256) and blends the result with the original weights, $\theta = \alpha\,\theta_{\text{FT}} + (1-\alpha)\,\theta_{\text{bge-m3}}$. **v2-base** and **v2-small** (e5-base, e5-small; scale 20, lr 3e-5) add a KL term to the softmax over in-batch scores of the frozen teacher (temperature 20, weight 1; teacher: the raw fine-tuned bge-m3). Selection used DEV only; TEST was scored for final candidates. The selection rules were written down before the runs they govern and amended twice; the notebook entries are quoted with timestamps in Appendix A. The first rule fixed α before any guard evaluation (within 0.5 dev macro points of the raw model and 1 point of bge-m3 on a mono-Arabic mMARCO proxy (Bonifacio et al., 2021)), which selected α = 0.75. Confidence intervals are percentile bootstraps over queries (2,000 resamples, within slices), paired when comparing systems.

5 Results

5.1 The asymmetry and how far fine-tuning closes it

Table 2 shows the test results and Table 3 the paired differences. Fine-tuning bge-m3 on the cross-lingual data and blending the weights back (v2-m3) lifts ar→en by +10.1 points and en→ar by +19.6; the macro average moves from 74.8 to 84.4. This closes the ar→en gap to English-to-English retrieval only partly: v2-m3 scores 73.5 against 90.0. Distillation transfers part of the gain: with the same data, a student trained without the teacher scores 60.7 (base) and 58.0 (small) on ar→en, and

Model	ar→ar	en→en	ar→en	en→ar	macro	95% CI (macro)
mE5-small	76.8	87.8	41.4	61.7	66.9	[66.0, 67.9]
v1 (e5-small FT)	80.2	87.3	55.1	80.4	75.7	[74.9, 76.6]
e5-small FT, no KD	79.4	86.4	58.0	80.2	76.0	[75.2, 76.9]
v2-small (KD)	80.8	87.6	60.2	81.1	77.4	[76.6, 78.3]
e5-base FT, no KD	78.9	86.3	60.7	81.4	76.8	[76.0, 77.7]
v2-base (KD)	81.8	88.6	64.5	84.1	79.8	[78.9, 80.6]
bge-m3	80.2	87.8	63.3	67.8	74.8	[73.9, 75.7]
v2-m3 raw FT	**87.1**	90.0	**73.9**	**88.4**	**84.8**	[84.1, 85.6]
v2-m3 (WiSE 0.75)	86.7	**90.0**	73.5	87.3	84.4	[83.6, 85.1]
mE5-base (not fine-tuned)	[TODO: baseline not yet scored on the TEST split]					

Table 2: Test MRR@10 ×100 per direction. Models are grouped by size (118M, 278M, 568M). Bold: best of the column. v1 is seed 13 (the three-seed mean of v1 is lower in ar→en); "no KD" rows use the same data as the distilled model.

	Δ ar→en		Δ macro	
Comparison (B vs A)	pts	95% CI	pts	95% CI
v1 vs mE5-small	+13.7	[+11.8, +15.6]	+8.8	[+8.1, +9.6]
v2-small vs v1	+5.1	[+3.9, +6.3]	+1.7	[+1.3, +2.1]
KD effect, small	+2.2	[+1.3, +3.1]	+1.4	[+1.1, +1.7]
KD effect, base	+3.8	[+2.6, +5.0]	+2.9	[+2.5, +3.3]
v2-m3 vs bge-m3	+10.1	[+8.7, +11.6]	+9.6	[+9.0, +10.3]
WiSE cost (raw to 0.75)	−0.4	[−1.0, +0.2]	−0.4	[−0.7, −0.2]
v2-base vs bge-m3	+1.2	[−0.9, +3.1]	+5.0	[+4.2, +5.8]

Table 3: Paired differences on the test split (percentage points).

with it 64.5 and 60.2, far below the teacher's 73.9. The 278M v2-base is statistically indistinguishable from the off-the-shelf 568M bge-m3 on ar→en (the paired interval includes zero) and ahead of it on the macro average.

5.2 Data scaling and the translated-query offset

Table 4 shows dev results for the e5-small student as translated training queries are added. The first 2,642 new translations give +1.5 dev ar→en points, 59% of the gain obtained with the 10,568 available at that time (+2.6); the curve flattens quickly. The e5-base student gains +1.9. The last row of each block uses the contamination-filtered full set of 19,886 new translations. On the test split the same kind of extra data did not help the small students and helped bge-m3 (Table 5), so the curve should be read as saturating, not as a recipe for more translations. Dev cross-lingual queries are machine-translated and score higher than human questions (Section 3), so only the differences between rows are meaningful.

DEV; training queries	new transl.	ar→en	en→ar	macro
e5-small student (no KD)				
none (v1 translations only)	0	65.2	77.9	77.0
+25% of the new translations	2,642	66.7	77.9	77.6
+first batch of new translations	3,904	66.9	79.1	77.8
+50% of the new translations	5,284	66.7	79.5	77.6
+all translations available at the time	10,568	67.8	79.7	77.8
full set, contamination-filtered	19,886	68.0	80.0	78.4
e5-base student (no KD)				
none (v1 translations only)	0	68.7	81.4	78.9
+all translations available at the time	10,568	70.6	82.1	79.3
full set, contamination-filtered	19,886	71.0	81.9	78.8

Table 4: Data-scaling curve (DEV, no distillation); "new transl." counts the new translated queries in each run's training set. Dev cross-lingual queries are machine-translated.

	ar→en		macro	
Student	partial	full	partial	full
e5-small	56.7	58.0	76.2	76.0
e5-base	59.9	60.7	77.3	76.8
bge-m3	72.6	73.9	84.2	84.8

Table 5: Test MRR@10 ×100 with the translated queries available at the first runs (partial) versus the full v2 set.

5.3 Negative results

Table 6 lists changes that did not help. Of the 11 dev comparisons none raises dev macro MRR@10 by more than +0.3 points (the largest drop is −2.4); baselines are 77.0 macro and 65.2 on ar→en for the small student, and 85.3 and 81.0 for the plain fine-tune of bge-m3 (held-out MIRACL-ar slice: 79.1). Distilling from the off-the-shelf bge-m3 does not help the small student, whereas distilling from the fine-tuned model does (Table 2): the teacher's cross-lingual skill is what is transferred. Replay of MIRACL-ar training questions, a preservation loss, and freezing parts of the encoder did not improve the held-out MIRACL-ar slice beyond what weight

Change (DEV, versus its own baseline)	Δ macro	Δ ar→en	Δ MIRACL-ar	kept
Other-language hard negative	+0.1	+0.5	–	no
Two same-language hard negatives	+0.1	+0.5	–	no
Language-homogeneous batches	−0.1	−0.1	–	no
Symmetric InfoNCE	−0.4	−0.7	–	no
KD from off-the-shelf bge-m3, w=1	0.0	+1.0	–	no
KD from off-the-shelf bge-m3, w=3	−1.0	−0.2	–	no
KD from off-the-shelf bge-m3, KD only	−2.4	−1.5	–	no
Two epochs	+0.3	+0.6	–	no
MIRACL-ar replay	−0.4	−0.2	−0.4	no
Embedding preservation to frozen bge-m3	−0.4	−0.1	−0.1	no
Frozen embeddings + bottom 12 layers	−0.3	+0.1	−0.3	no
LoRA (r=32)	[TODO: run pending]			no
Dynamic int8 export, v2-small / v2-base (TEST)	−2.4 / −5.4	−5.1 / −8.3	–	no

Table 6: Negative and null results, effect on dev versus the baseline of the same group (small student: no change; bge-m3: plain fine-tune); the last column says whether the change is in the final recipe. Δ MIRACL-ar: held-out slice of MIRACL-ar training questions, selection only.

DEV selection set	ar→en	macro	held-out MIRACL-ar
e5-base student (teacher: raw FT bge-m3)			
no KD	71.0	78.8	–
KD w=1	74.5	81.2	–
KD w=3	75.6	81.8	73.1
KD only	76.1	82.0	73.2
bge-m3 fine-tuned itself			
bge-m3 (no FT)	73.5	78.2	80.5
raw FT	81.0	85.3	79.1
WiSE α=0.75	80.5	84.9	79.8
WiSE α=0.50	79.8	84.2	79.8
bge-m3 + MIRACL-ar replay / preservation			
replay	80.8	84.9	78.7
preservation	80.9	84.9	79.0
preservation, WiSE 0.75	80.6	84.7	79.8

Table 7: DEV MRR@10 ×100 for distillation and weight-interpolation variants; the last column is the held-out MIRACL-ar slice.

interpolation gave. Dynamic int8 export (Section 6) is included for completeness.

5.4 The WiSE trade-off curve

Interpolating the fine-tuned and original bge-m3 weights moves the model along a curve between the original model ($\alpha = 0$) and the raw fine-tune ($\alpha = 1$). Figure 1 plots test ar→en, test macro and the guard tasks against α. The curve has [TODO: pending] points; at the selected α ([TODO: pending]), ar→en is [TODO: pending] and MIRACL-HN [TODO: pending], against [TODO: pending] for bge-m3. The largest ar→en that keeps MIRACL-HN within 1 point of bge-m3 is [TODO: pending], at α = [TODO: pending]. **The curve is post hoc**: it was computed after α had been selected, so it describes the trade-off but did not choose the operating point. [TODO: WiSE curve: alpha_curve.json pending]

[TODO: WiSE curve: results/retrieval_v2/alpha_curve.json pending; report/make_figures.py draws it]

Figure 1: WiSE trade-off (post hoc): fine-tuned share α against test ar→en, test macro and the guard tasks; the circle is bge-m3, the dashed line the selected α.

5.5 MTEB(ara, v1) and its uncertainty

bge-m3 leads the leaderboard at 72.0 mean over tasks, and the v1 fine-tune (65.0) ends below its own starting point, multilingual-E5-small: the cross-lingual gain was paid for on tasks unlike the training data, in particular Mr.TyDi re-ranking (79.2 to 65.8) and MIRACL (72.4 to 65.8). With 19 tasks, differences of a point or two are within noise; Table 9 quantifies this with a bootstrap over tasks and an exact sign test. v2-m3 and bge-m3 differ by +0.06 mean points (95% interval −1.1 to +1.1), with 9 wins, 9 losses and 1 tie over 19 tasks (sign test p = 1.00): the aggregate cannot separate them, because gains on retrieval tasks (largest: MLQARetrieval, +4.0) are offset by losses on MIRACL and Mr.TyDi re-ranking (largest: NamaaMrTydiReranking, −7.3). On the Arabic↔English subsets alone, v2-m3 is ahead by +2.7 (+1.1 to +4.3) with 5 wins, 0 losses and 1 tie over 6 tasks (sign test p = 0.06), consistent with the gains on the held-out benchmark but based on very few tasks. For scale, bge-m3 against multilingual-E5-base differs by +4.1 (+2.8 to +5.4, p = 0.001), so the benchmark does resolve gaps of that size. Students against multilingual-E5-base: [TODO: pending] ([TODO: pending] to [TODO: pending]).

5.6 Guard tasks and the contamination analysis

Table 10 reports three guard tasks in the MTEB protocol. Relative to bge-m3, v2-m3 is +2.7 points on SemRel24, but −3.3 on MIRACL-HardNegatives (75.7 vs 79.0) and −7.3 on Mr.TyDi re-ranking. Our bar was to stay within 1 point of bge-m3 on MIRACL while keeping test ar→en at 72 or above; it is *not met*, and neither replay nor preservation closed the gap. Contamination does not explain the gap away: splitting the queries into near-duplicates of TyDi-ar or MIRACL-ar training questions and the rest (Table 11), the difference between v2-m3 and bge-m3 on the clean MIRACL queries is −4.4 points (all queries: −5.4). Only 184 MIRACL queries are clean, so the interval is wide, and the

Model	Task	Type	Retr.	Rerank	STS	Cls.	ar	ar-en
bge-m3	72.0	70.9	73.5	79.4	66.3	64.5	70.4	73.3
multilingual-e5-large	71.4	70.3	74.0	79.6	68.1	61.8	69.5	75.3
multilingual-e5-base	67.9	67.3	68.8	77.1	64.9	59.6	66.5	69.1
multilingual-e5-small	65.6	65.7	66.0	79.2	60.1	58.3	65.0	63.9
arabic-english-e5-small-retrieval	65.0	64.4	66.3	65.9	57.4	59.0	63.1	68.1
paraphrase-multilingual-MiniLM-L12-v2	58.7	62.7	45.2	70.0	63.1	52.0	54.6	66.5
silma-embeddding-matryoshka-v0.1	60.3	61.9	47.1	71.4	59.2	63.2	62.2	47.9
LaBSE	58.2	58.7	46.2	51.6	62.5	58.3	54.2	66.2
Arabic-Retrieval-v1.0	58.3	56.8	55.8	65.2	62.8	64.0	67.4	27.9
Arabic-Triplet-Matryoshka-V2	55.7	54.8	51.0	64.5	60.3	63.4	66.3	19.5
v2-m3	72.0	70.4	74.3	72.1	67.8	64.1	70.0	75.9
v2-base	[TODO: pending]							
v2-small	[TODO: pending]							

Table 8: MTEB(ara, v1), scores $\times 100$. Task = mean over tasks, Type = mean over task types; ar / ar-en = mean over Arabic-only / Arabic–English subsets. v2 rows are recomputed from the per-task scores of the paired comparison.

Mean (Task) difference, new − ref	tasks	diff	95% CI (tasks)	W/L/T	sign p
v2-m3 vs bge-m3	19	+0.1	[−1.1, +1.1]	9/9/1	1.00
ar-en cross-lingual subsets	6	+2.7	[+1.1, +4.3]	5/0/1	0.06
ar-only subsets	17	−0.4	[−1.6, +0.7]	7/10/0	0.63
v2-base vs mE5-base (student)	[TODO: MTEB(ara) scoring of the students pending]				
reference pairs from the 10-model leaderboard					
bge-m3 vs mE5-base	19	+4.1	[+2.8, +5.4]	17/2/0	0.001
mE5-base vs mE5-small	19	+2.3	[+0.9, +3.7]	14/5/0	0.06
v1 vs mE5-small	19	−0.6	[−2.9, +1.4]	10/9/0	1.00
bge-m3 vs mE5-large	19	+0.5	[−0.5, +1.6]	11/8/0	0.65

Table 9: Paired uncertainty of MTEB(ara) mean differences: bootstrap over tasks, wins / losses / ties, exact two-sided sign test. The lower block compares models of the 10-model leaderboard, for scale.

Model	MIRACL-HN	Mr.TyDi rerank	SemRel24
mE5-small	72.4	79.2	51.0
v1	65.8	65.8	41.2
mE5-base	73.0	77.2	59.2
bge-m3	79.0	79.4	59.1
v2-m3 (WiSE 0.75)	75.7	72.1	61.8
v2-m3 − bge-m3	−3.3	−7.3	+2.7

Table 10: Guard tasks (MTEB protocol, $\times 100$): MIRACL-HardNegatives (nDCG@10), Mr.TyDi reranking (MAP), SemRel24 (Spearman). These runs use slightly different numerics than Table 8 (differences up to 0.5 points for the same model and task).

split used a slightly different scoring input than the MTEB protocol. [TODO: split scores are 3.4 pts below the MTEB protocol for bge-m3 (titles missing); rerun pending] bge-m3 lists MIRACL and Mr.TyDi among its training sets, so its absolute MIRACL number is not a clean reference either.

5.7 Confirmatory seeds

The model-selection experiments above used the first three seeds and several amended selection rules, so their headline numbers are exploratory (Table 12). Two further seeds ([TODO: seeds 16 and 17 pending]) were trained after the recipe was frozen ([TODO: freeze timestamp: quote from RESULTS_v2.md]); Table 13 is the confirmatory result. Their mean ar→en is [TODO: pending] (macro [TODO: pending]), [TODO: pending] and [TODO: pending] points relative to bge-m3.

6 Deployment Costs

We export v2-small and v2-base to ONNX (fp32, opset 17, pooling and normalisation inside the graph) and compare three ways of shrinking them for CPU serving: ONNX Runtime dynamic int8 with per-tensor or per-channel weight scales (activations quantised on the fly, cf. Zafrir et al., 2019), and weight-only 8-bit block quantisation of the MatMul weights that leaves the activations in fp32. Quality is measured as MRR@10 on the same test split against the PyTorch fp32 model, on CPU, with paired bootstrap intervals. Latency was measured on the 20-core ARM CPU of a shared server (not an x86 laptop or cloud VM), one process, 2 and 8 threads; numbers are therefore indicative.

Table 14 shows that the ONNX fp32 graphs reproduce the PyTorch results (0.00 and 0.00 macro points) and cut single-query latency (2 threads) from 18 to 6 ms for v2-small and from 48 to 32 ms for v2-base. Dynamic int8 with per-tensor scales is faster still (4 and 7 ms) but *not acceptable* here: −2.44 macro points for v2-small and −5.42 for v2-base (−8.27 on ar→en for the base model). Per-channel scales reduce the loss to −0.72 and −4.56, still more than we would ship for the base model. Weight-only 8-bit quantisation is lossless within noise (+0.04 and −0.11) and shrinks the models from 448 to 114 MiB and from 1059 to 270 MiB, but its CPU kernels are slower per query than fp32

Model	MIRACL-HN (n=184+816) all	clean	contam.	Mr.TyDi rerank (n=894+24) all	clean	contam.
bge-m3	75.7	73.1	76.2	79.4	79.1	89.2
mE5-base	70.4	67.3	71.1	77.4	77.2	82.6
v1	64.0	61.8	64.5	65.8	65.6	72.5
v2-m3 raw FT	66.8	64.9	67.3	69.4	69.3	73.8
v2-m3 (WiSE 0.75)	70.2	68.8	70.6	72.1	71.9	79.3

Table 11: Guard tasks split by contamination: "clean" queries are not near-duplicates of any TyDi-ar or MIRACL-ar training question, "contam." queries are. Per-query scores in the contamination-split protocol.

TEST	ar→en	macro
v2-m3 raw FT, seed 13	73.9	84.8
v2-m3 raw FT, seed 14	73.2	84.3
v2-m3 raw FT, seed 15	72.7	84.3
mean ± s.d. (3 seeds)	73.3 ± 0.6	84.5 ± 0.3
v2-m3 WiSE 0.75, seeds 14 and 15	[TODO: blends pending]	
v2-base / v2-small, seeds 14 and 15	[TODO: runs pending]	

Table 12: Exploratory seeds of the raw fine-tune (test); pending runs are marked.

Recipe, seed	ar→en	macro	Δ ar→en vs bge-m3
seeds 16 and 17	[TODO: trained after the recipe was frozen; pending]		

Table 13: Confirmatory seeds, trained after the recipe was frozen (test). Intervals: paired bootstrap over queries against bge-m3.

(18 and 97 ms). We therefore recommend ONNX fp32 when latency matters, weight-only 8-bit when download size matters, and no dynamic int8 for these models without calibration or quantisation-aware training.

7 Data Card and Licences

Table 15 lists every dataset and model used, how it entered the study, and the licence on its Hugging Face card (checked 2026-11-04). The final recipe trains on TyDi QA (Apache-2.0), SQuAD v1.1 (CC BY-SA 4.0) and ARCD (MIT) and on machine translations of their questions produced by Gemma-4-26B-A4B (Apache-2.0); the starting weights are bge-m3 (MIT) and multilingual-E5 (MIT). MIRACL Arabic (Apache-2.0) was used only in an ablation, XQuAD (CC BY-SA 4.0) only for testing, and the remaining sources only for validation or selection. Passage text is Wikipedia content, so share-alike terms of SQuAD and Wikipedia apply to derived corpora. [TODO: licence of the released weights: author decision; the training data include CC BY-SA sources]

8 Limitations

Most fine-tuned numbers are single seeds; the confirmatory seeds cover the final recipe only, and the WiSE blends and students of other seeds are pending (Tables 12, 13). The test set is Wikipedia-style extractive QA with 1,190 parallel questions per XQuAD direction, so the intervals are wide for the cross-lingual slices; dialects, informal text and long documents are untested. Training queries are machine-translated and not human-verified; the offset check cannot rule out subtler translationese, and dev cross-lingual numbers are inflated by it. The guard analysis shows a real regression on MIRACL and Mr.TyDi, and the clean subset is small. MTEB(ara, v1) has 19 tasks and 10 scored models; its aggregate cannot resolve differences of about a point (Table 9). The selection rules were amended during the work (Appendix A); the post-hoc WiSE curve and the exploratory seeds should not be read as confirmatory. Deployment latency comes from one machine type, with other jobs running, and does not include serving overhead. No RAG-level evaluation is done here.

9 Reproducibility

Every number in this report is generated by `report/make_tables.py` from the committed result files in `results/retrieval_v2`, `results/mteb_ara` and `deploy/results`; the figure by `report/make_figures.py`; run `make -C report pdf`. Training used a single GB10 GPU; the evaluation harness is in the accompanying repository and the deployment scripts are in `deploy/`. The data are public (Table 15); the v2 model weights are [TODO: published after review: Hugging Face repository names].

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Variant (CPU)	MiB	RSS	q@1 ms	p/s	MRR	Δ	Δ ar→en
v2-small PyTorch fp32	449	1818	18	14	77.4	ref.	ref.
v2-small ONNX fp32	448	2613	6	20	77.4	0.00	0.00
v2-small ONNX int8	113	2249	4	34	75.0	−2.44	−5.09
v2-small ONNX int8, per-channel	113	2258	4	33	76.7	−0.72	−1.68
v2-small ONNX weight-only 8-bit (MatMulNBits)	114	2266	18	20	77.5	+0.04	+0.05
v2-base PyTorch fp32	1061	2447	48	3	79.7	ref.	ref.
v2-base ONNX fp32	1059	3089	32	4	79.7	0.00	0.00
v2-base ONNX int8	265	2708	7	11	74.3	−5.42	−8.27
v2-base ONNX int8, per-channel	266	2736	9	11	75.1	−4.56	−6.50
v2-base ONNX weight-only 8-bit (MatMulNBits)	270	2226	97	4	79.6	−0.11	−0.18

Table 14: CPU deployment variants. MiB: size on disk; RSS: peak resident memory of the benchmark process (8 threads; includes the python and data-loading baseline of that process); q@1: p50 latency of a single query (2 threads, ms); p/s: passages per second (batch 32, 8 threads); MRR: test macro MRR@10 ×100; Δ: difference to PyTorch fp32 in points (macro, and ar→en).

Source	Role	Licence (Hub card)
TyDi QA GoldP (Arabic, English)	training data (final recipe)	apache-2.0
SQuAD v1.1 train	training data (final recipe)	cc-by-sa-4.0
ARCD	training data (final recipe)	mit
XQuAD	evaluation / selection only	cc-by-sa-4.0
MIRACL Arabic (queries, qrels, corpus)	training data (ablation only)	apache-2.0
Gemma-4-26B-A4B (query translator)	translated the training queries	apache-2.0
bge-m3 (base weights of v2-m3, teacher baseline)	initial weights	mit
multilingual-E5-base (base of v2-base)	initial weights	mit
multilingual-E5-small (base of v2-small)	initial weights	mit
ArSarcasm (selection suite)	evaluation / selection only	mit
XNLI (selection suite)	evaluation / selection only	no licence tag on the Hub card
Arabic STS-B (selection suite)	evaluation / selection only	apache-2.0
mMARCO Arabic (selection proxy)	evaluation / selection only	no licence tag on the Hub card
Weights of v2-m3, v2-base, v2-small	released models	[TODO: licence of the weights: author decision (training data include share-alike sources)]

Table 15: Data card: sources, roles and licences. "No licence tag" means the Hub card does not state one; nothing is assumed.

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A Pre-registration Trail

The selection of the v2-m3 recipe followed rules that were written down in the lab notebook of the training machine before the runs they govern. Below, the notebook entries are quoted verbatim apart from LaTeX escaping, with their timestamps: the rule that fixed α, the original replay/preservation rule, amendment 1 (a validation suite, written before any candidate was scored on it, with a same-day correction) and amendment 2 (an extended range of α and a rerun of the LoRA candidate after a bug in the first run). The amendments changed the selection procedure after earlier selections had been made, which is why the confirmatory seeds and the post-hoc curve are reported separately.

WiSE alpha selection rule, as recorded in the results entry (2026-11-04 11:05) Selection rule fixed BEFORE any guard eval: dev macro within 0.5 of raw AND proxy within 1.0 of bge-m3 → alpha 0.75.

Original pre-registered rule (2026-11-04 14:00)
2026-11-04 ~14:00 coordinator: fix the MIRACL forgetting properly (replay + preservation), PRE-REGISTERED selection rule
- Why: v2-m3 + WiSE a.75 scores MIRACL-HN.v2 75.7 vs bge-m3 79.0 (-3.3); bar = within 1 pt while TEST ar→en >= 72.
- Replay data: MIRACL-ar TRAIN (miracl/miracl topics+qrels, miracl/miracl-corpus; Apache-2.0; judged-relevant = positives, judged-NOT-relevant = hard negatives), 500 queries HELD OUT as the selection-only slice devM (own mini-corpus of judged passages + 10k random distractors, evaluated separately, never in the dev macro). Filters: near-duplicate of any MIRACL-ar dev query / v1 test+dev query, positive overlapping a MIRACL dev positive or a v1 test passage; contam_v2 then applies the whole MTEB(ara) list. Mr.TyDi-ar train: not added (its questions ARE the TyDi-ar train questions v1/v2 already use; script-based loader).
- Preservation loss: 1 - cos(student, frozen bge-m3) on Arabic positives, weight 0.5 (sweep not affordable: one m3 run = 55 min).
- Runs (seed 13, m3 recipe of m3_b_s13): m3_r (replay), m3_p (preservation), m3_rp (both); each also WiSE a.75 / a.50 against bge-m3; evaluation = dev (4 directions), devM, mMARCO proxy. NO guard is used for selection.

- RULE (fixed now, before the runs): a candidate qualifies if (i) devM MRR@10 >= bge-m3's devM - 1.0, (ii) dev ar→en >= 79.0 (~ TEST 72), (iii) dev macro >= m3_b dev macro (85.3) - 1.5; among qualifying take the highest dev macro (ties: fewer moving parts). If none qualifies: highest devM among those with dev ar→en >= 79.0, reported as a miss.
- Seed 14 of the OLD recipe is allowed to finish (running); seed 15 is NOT run with the old recipe. KD ablations/baseline guards are queued after these runs.

Amendment 1: validation-suite selection (2026-11-04 19:05)

2026-11-04 ~19:05 -
PRE-REGISTRATION: validation suite for the final v2-m3 pick (written BEFORE any candidate is scored on the new components)
Trigger: coordinator, "0.06 is a tie; the losses (Namaa, MIRACL, XNLI, AJGT) are drift away from bge-m3's general skills; select on a validation suite that mirrors the MTEB(ara)
task types, never on MTEB". Supersedes select_m3.py (kept as the record of the earlier rule). No MTEB(ara) number and no TEST number is read by this procedure.
Order of work: s15 → soup/blends → suite (interim table) → LoRA / freeze → suite (all) → selection → ONE MTEB(ara) scoring of the pick → teacher = pick → students.
DEVIATION FROM THE BRIEF (flagged to the coordinator): the brief asks for the XNLI-ar VALIDATION split "since MTEB uses test". mteb's XNLI task sets eval_splits=["test","validation"] (pylibs/mteb/tasks/pair_classification/multilingual/xnli.py) and the leaderboard number is the mean over both, so the validation split is
part of the MTEB evaluation. Instead: a seeded 3,000-pair subsample of the XNLI-ar TRAIN split (facebook/xnli, ar/train; machine-translated MNLI; entailment=1, contradiction=0, neutral dropped, as mteb does). Never trained on, never part of any MTEB split.
COMPONENTS (6, all x100, higher = better):
1 dev : v2 extended DEV macro MRR@10 over the 4 directions (ar→en, en→ar, ar→ar, en→en) [results/NAME.dev.json]
2 devM : MIRACL-ar TRAIN hold-out (500 queries, selection-only slice, excluded from training), MRR@10 [NAME.devM.json]
3 proxy : mMARCO-ar dev proxy (1,000 q / 25k docs, seed 13), MRR@10 [results/ood_dev_ar.json]
4 xnli : average precision of the cosine similarity on 3,000 XNLI-ar TRAIN pairs (seed 13) [results/suite/NAME.json]
5 cls : mean of two 8-shot-per-label logistic-regression probes (mteb protocol: 10 experiments, samples_per_label=8, max_iter=100) on ArSarcasm
(iabufarha/ar_sarcasm, licence MIT): (a) sarcasm 2-class, (b) sentiment 3-class. The probe is fitted on rows of the TRAIN split and scored on
2,000 other TRAIN rows (seeded hold-out). MTEB scores TweetSarcasmClassification on the ar_sarcasm TEST split; the TRAIN split is disjoint
from it (same source, so it is a close proxy by design). AJGT / TweetEmotion / HotelReview are not touched.
6 sts : Spearman of cosine similarity on the Arabic DEV split of stsb_multi_mt (PhilipMay/stsb_multi_mt, ar/dev, 1,500 pairs; machine-translated STSb; licence
"other" - used for validation only, not redistributed). Disjoint from STS17 / STS22 / SemRel24 (the MTEB STS tasks).
NORMALISATION: per component c, x_c is min-max scaled over the POOL = {all candidates + the reference rows (bge-m3, m3_b_s13 raw, soup raw)}:
n_c = (x_c - min_pool) / max(max_pool - min_pool, 1.0 point) (the 1-point floor keeps a component whose pool range is below the noise level from being inflated)
SCORE = unweighted mean of the 6 n_c. Reported with the 6 raw components.
POOL / CANDIDATES (selectable): WiSE alpha in {0.4,0.5,0.6,0.75} of m3_b_s13 (wm_a50, wm_a75 exist; w_m3b_a40, w_m3b_a60 new);
SOUP = uniform average of the seed 13/14/15 fine-tunes (m3_b_s13/s14/s15; seed 15 is trained first), then WiSE alpha in {0.4,0.5,0.6,0.75} (soup_a40..a75);
LoRA r=32 attention+MLP lr 1e-4 (m3_lo_s13) and freeze-bottom-12 (m3_fr_s13) raw, plus their WiSE blends alpha {0.75,0.5}. 14 selectable models.
Reference rows (in the pool for normalisation, shown, NOT selectable): bge-m3, m3_b_s13 raw (alpha=1), soup raw.
ELIGIBILITY (protects the S_TIER ar→en bar): dev ar→en >= bge-m3 dev ar→en + 5.0 points.
PICK = the eligible candidate with the highest SCORE. Ties (|diff| < 0.02) → fewer ingredients: single-seed WiSE < soup WiSE < LoRA/freeze (then larger alpha).
The pick is announced here before its MTEB(ara) scoring; that scoring is run once and reported even if it is lower than final_m3_wa75_s13 (72.044). Earlier MTEB(ara)
scoring of final_m3_wa75_s13 happened BEFORE this suite existed and is not used in the selection.
3-SEED REPORTING: pick is a soup → the soup is the reported model (one model; paired bootstrap over queries); otherwise the per-seed blends (13/14/15) of the picked recipe.
Students: distilled from the FINAL teacher (the pick), KD-only; teacher embeddings are recomputed from the pick (tag m3f).
Code: lab/retrieval/v2/suite_v2.py (components 4-6), suite_select.py (assembly + rule), runs/retrieval/v2/scripts/drive14.sh (queue).
- 2026-11-04 ~19:20 CORRECTION to the pre-registration above (still before any candidate has a suite number): PhilipMay/stsb_multi_mt has NO Arabic split (checked on the Hub),
so component 6 uses the Arabic STS-B VALIDATION split of Omartificial-Intelligence-Space/Arabic-stsb (apache-2.0; machine-translated STS-B, 1,500 pairs; validation.csv, "score" column).
Everything else is unchanged. Data fetched to runs/retrieval/v2/data/suite/ (xnli_ar_train.parquet, ar_sarcasm_train.parquet, stsb_ar_validation.csv).

Amendment 2: alpha extension and LoRA rerun (2026-11-04 21:35)

2026-11-04 ~21:35 -
AMENDMENT 2 to the pre-registered suite selection (coordinator follow-up after DRIVE14_DONE); written BEFORE any of the runs below
What happened in drive14 (facts, from the logs; the "rc=0" lines of the driver are NOT reliable because run_docker.sh masks the exit code - from now on every step is verified by the existence of its output file):
* LoRA: m3_lo_s13 trained all 615/615 steps (56 min) and then CRASHED in train_v2.py at merge_and_unload (AttributeError: 'XLMRobertaModel' has no attribute 'merge_and_unload'): my code
kept the in-place-injected XLM-R module instead of the PeftModel handle. Nothing was saved (the adapters died with the process), so the LoRA dev / merges / blends / proxy2 / suite2 all failed on the missing path
within seconds. This is a bug in MY code, not a result about LoRA. proxy2 and suite2 aborted at their FIRST model (LoRA), so the freeze run and its two blends (which trained fine: dev 85.0 / 84.8 / ..., devM 78.8 / 79.7 / ...) have no proxy and suite numbers.
* select2's "PICK (interim)" was computed on the 11 complete models only.
* Tie rule: the pre-registration says "|diff| < 0.02" (strict). select code used >= best - 0.02 (inclusive). Corrected to strict;

exact differences are printed to 4 decimals; the pick report states which clause decided.
DECISIONS (no re-litigation after this entry):
A. LoRA is RERUN with the identical recipe / seed (m3_lo_s13, r=32, lr 1e-4, attention+MLP, 1 epoch, seed 13) after a gated 3-minute merge self-test of the fixed code (adapter saved to disk before the merge). ~57 min. Reason: it is the only
pre-registered candidate that constrains the update DURING training (freeze did not help on devM); the GPU would otherwise idle until the pick. It enters the pool exactly as registered (raw + WiSE .75 / .5).
B. Order: (1) freeze proxy + suite → (2) select on the pool WITHOUT LoRA → winning family F by the rule as written (strict tie) → (3) ONE extra pair of alphas for F ONLY: alpha 0.30; alpha 0.35 is added ONLY if the
0.30 model fails eligibility (dev ar→en < 78.5 = bge-m3 dev ar→en 73.5 + 5.0); if 0.30 is eligible, 0.35 is NOT run; nothing below 0.30 is tried → (4) LoRA rerun + blends → (5) FINAL select on the FULL pool
(all earlier candidates + the 0.30/0.35 point(s) + LoRA models), same normalisation (pool-wide min-max, 1-point floor), same eligibility, same strict tie rule → (6) ONE MTEB(ara) scoring of that pick (= final_m3), reported even if lower.
If LoRA wins the final select there is no further alpha extension. The alphas 0.30 / 0.35 are selectable. Family names for the extension: wise → w_m3b_a30 / a35; soup → soup_a30 / a35; freeze → w_m3_fr_s13_a30 / a35.
C. Everything else in the first pre-registration (components, normalisation, eligibility, tie-break by ingredients, 3-seed reporting) is unchanged. No MTEB(ara) and no TEST number is read by any of the selection steps.